

The Symbiotic Search Fabric

A Co-Adaptive, Multi-Coupled-Tenant, Client-Owned, Agent-Operated,
Time-Bounded Architecture for Search Systems Whose Design
Evolves from the Experience of Their Operating Agents

by

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Declaration

This manuscript is a **design synthesis**, prepared in the form and to the conventions of a master's thesis. Its subject — the Symbiotic Search Fabric — is a notional architecture: no measured implementation exists, and no claim in this document should be read as an empirical result. Wherever a constituent mechanism has an established basis in the published literature (anytime algorithms, differential privacy, secure aggregation, Shapley-value accounting, learning-coefficient estimation, market-based scheduling, provenance modelling), the text marks it *adopt, not invent* and cites the source; the architectural contribution claimed here lies in the synthesis, the interface discipline, and the design-space analysis, not in the constituent mechanisms.

All counts in Chapter 8 are illustrative arithmetic over a stated enumeration, not measurements. The evidentiary seam between what is designed, what is adopted, and what is speculated is preserved deliberately throughout, and the falsification criteria in Chapter 10 are offered so that the work can be refused honestly — with a counter-offer — by any examiner it fails to convince.

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Abstract

Autonomous agents increasingly outnumber people as the operators of search infrastructure, yet the architectures they inhabit remain designed around human querying habits: best-effort latency, custodial data ownership, isolated tenancy, and design change managed as an out-of-band engineering activity. This thesis asks what a search system looks like when five commitments are taken as constitutional rather than aspirational — agents operate it, clients own it, tenants are coupled through it, every interaction is time-bounded within it, and the lived experience of its operating agents is the raw material from which its own design evolves.

The thesis decomposes its originating brief into ten load-bearing obligations and discharges them with a complete architecture: four planes (a human charter plane that governs without operating; sovereign tenant estates with residence-time-stratified indexes, quarantined relevance learning, and unconditional experience ledgers; a graded coupling membrane that is consented, privacy-metered, and unilaterally revocable; and a protocol-only operator substrate) crossed by two transverse structures (a temporal contract spine enforcing deadline contracts with honest refusal, and a six-stage evolution pipeline through which distilled experience — and nothing else — becomes countersigned, versioned design). Relationships among components are specified once, in a closed-world interface registry of twenty-nine edges, and bound by a lattice of seven invariants. A phased implementation roadmap assigns each mechanism a validation burden and exit criterion. Finally, the design space itself is analysed: the ten obligations admit 589,824 raw stance-combinations, of which roughly one in five survives joint admissibility, collapsing under structural equivalence into seven load-bearing archetype families; a lexical selection rule — sovereignty and honest time prior to co-adaptation, co-adaptation prior to efficiency — is shown to select the proposed architecture and, permuted, to generate its rivals.

Keywords: agent-operated search; multi-tenant architecture; data sovereignty; federated learning; anytime algorithms; deadline scheduling; co-adaptive systems; design-space analysis; design science.

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Nomenclature

Symbol	Meaning
$\langle d, q_{\min}, b \rangle$	query contract: deadline, quality floor, budget (SP-1)
$\langle d', q' \rangle$	counter-offered contract accompanying a refusal (SP-4)
$\lambda(t)$	shadow price of shared capacity at time t (SB-2)
$\lambda\text{-hat}$	online drift / learning-capacity estimate on the relevance organ (ES-3a)
ϵ	privacy budget spent per membrane exchange; finite per consent (MB-3)
κ, κ_{AB}	coupling consent; the $A \leftrightarrow B$ instance (MB-2)
θ, θ_T	relevance-organ parameters; tenant T 's instance (ES-3)
τ	residence time of an artifact in a stratum (ES-2)
$S_0 / S_1 / S_2$	hot lattice / working index / deep archive (ES-2)
$R_0\text{--}R_3$	deadline rungs: reflex, interactive, deliberative, campaign (SP-2)
$G(n)$	the n -th generation frame; $G(n+1)$ is always compiling (EV-9)
$\ G \circ E\ $	co-adaptive loop gain; release requires a value below one (EV-7, I7)
$E[q \mid d, b]$	expected answer quality under a contract, estimated from ledger history (SP-3)
B	tenant budget envelope fixed in the charter (CH-2)
$P_1\text{--}P_{10}$	the ten obligations decomposed from the originating brief (Chapter 3)
$I_1\text{--}I_7$	the invariant lattice (Section 7.4)
$IF\text{-}01\text{...}IF\text{-}29$	edges of the closed-world Interface Registry (Table 7.1)
$CH / ES / MB / SB / SP / EV$	component prefixes: charter, estate, membrane, substrate, spine, evolution pipeline

1 · Introduction

This chapter motivates the problem, states the originating brief, formulates research questions and contributions, and fixes the methodological and evidentiary posture of the thesis.

1.1 Motivation

Search infrastructure is quietly changing operators. Retrieval-augmented and tool-using agents [21, 37] now issue queries at machine cadence, refine them without human pause, and consume answers as inputs to further computation rather than as pages to be read. Yet the systems they query remain shaped by human assumptions: latency is a service-level aspiration rather than a contract; tenancy is an isolation mechanism rather than a legal relationship [5]; the index belongs in practice to whoever hosts it, whatever the contract says; and the design of the system evolves through vendor release cycles that owe nothing, formally, to the experience of the agents who operate inside it. Each of these assumptions fails a different stakeholder — the agent that cannot plan around an unbounded tail latency, the client whose corpus and learned relevance models are hostage to an exit negotiation, the federation of tenants who could learn from one another but dare not share, and the operator who cannot explain why the system’s own evolution should be trusted.

The interesting question is not how to patch any one failure but what architecture results when their negations are taken as first principles. That is the question this thesis answers by construction.

1.2 Problem statement and originating brief

The work originates in a single design brief, quoted verbatim because Chapter 3 holds itself answerable to it clause by clause:

“Envision a mature end-to-end framework for co-adaptive, multi-coupled-tenant, client-owned, agent-operated, time-bounded search systems where agents’ experience shapes the evolving design.”

The problem, precisely stated: **specify a complete architecture that discharges every commitment in this sentence jointly** — not a menu of mechanisms from which some subset might be assembled, but one coherent system in which the commitments reinforce rather than undermine one another,

presented with sufficient rigor that its interfaces are unambiguous and its claims falsifiable.

1.3 Research questions

RQ1. What set of atomic obligations does the brief actually impose, and can each be given a precise design question? (Chapter 3.) **RQ2.** What architecture discharges all obligations jointly, and how can its components and their relationships be specified with minimal ambiguity? (Chapters 4-7.) **RQ3.** How large is the space of alternative syntheses of the same obligations, what structure does it have, and what selection rule justifies choosing this architecture over its rivals? (Chapter 8.) **RQ4.** By what programme could the design be implemented and validated, and what observations would falsify it? (Chapters 9 and 10.)

1.4 Contributions

C1 — A requirements decomposition. The brief is decomposed into ten load-bearing obligations (P1-P10), each with its design question and a traceability map to the architecture (Chapter 3). **C2 — A complete architecture.** Four planes and two transverse structures, specified component-by-component, with relationships confined to a closed-world interface registry of twenty-nine edges and a lattice of seven invariants (Chapters 4-7). **C3 — A design-space analysis.** An explicit stance enumeration over the ten obligations (589,824 raw combinations), six compatibility rules, an equivalence quotient yielding seven archetype families, and a lexical selection functional that generates both the chosen architecture and its rivals (Chapter 8). **C4 — A validation programme.** A five-phase implementation roadmap in which every mechanism carries a validation burden and exit criterion, together with four concrete falsifiers (Chapters 9-10).

1.5 Methodology and evidentiary posture

The thesis follows the design-science research paradigm [15, 29]: the artifact is the contribution, and its evaluation here is analytical and programmatic — internal-consistency argument, design-space comparison, and a pre-registered validation plan — rather than empirical, since no implementation exists. Three disciplines govern the writing. First, components are introduced **before** any relationship among them is asserted: Chapters 5 and 6 name every part and state its function in isolation; Chapter 7 is the only place where edges are defined, under a closed-world convention in which absence from the registry is prohibition. Second, every mechanism carries its evidentiary status; the marker *adopt, not invent* separates the synthesis (claimed) from its ingredients (cited). Third, the design treats its own decisions as contingent: Chapter 8 counts the roads not taken, in the spirit of documented — rather than idealized — design rationale [28, 3].

1.6 Thesis outline

Chapter 2 situates the work in six bodies of literature. Chapter 3 decomposes the brief into obligations P1-P10. Chapter 4 states the governing rule and the macro-architecture. Chapter 5 treats each plane in dedicated detail; Chapter 6 does the same for the two transverse structures. Chapter 7 specifies all relationships and the invariant lattice. Chapter 8 analyses the design space of alternative syntheses. Chapter 9 gives the implementation roadmap and validation programme. Chapter 10 discusses findings, falsifiability, and limitations; Chapter 11 concludes.

2 · Background and Related Work

Six bodies of literature bear on the design. This chapter reviews each and states the gap the thesis occupies: not a missing mechanism, but a missing synthesis.

2.1 Search and retrieval under time bounds

Classical information retrieval provides the substrate vocabulary of indexes, ranking, and evaluation [2]; modern approximate-nearest-neighbor structures such as HNSW [23] provide graph indexes whose descent can be truncated early — the mechanical property this thesis elevates into law. The idea that a computation should return the best answer available at interruption, with quality monotone in time, is the anytime-algorithm tradition [6, 38]; hard admission of work against deadlines descends from real-time scheduling [22]. What the literature treats as performance engineering, this thesis treats as contract law: Section 6.1 binds anytime kernels and deadline classes into an admission discipline with an explicit refusal protocol.

2.2 Agents as operators

Multi-agent systems provide role decomposition, negotiation, and coordination theory [36]; the blackboard tradition [9] is the canonical shared-workspace synthesis that Chapter 8 revisits as an archetype. Contemporary language-model agents interleave reasoning with tool calls and retrieval [37, 21], making machine-cadence search operation a present reality rather than a projection. Autonomic computing anticipated systems that manage themselves under human policy [18]; the thesis sharpens that vision’s division of labour into a constitutional one: humans hold charters and countersignatures, agents hold every operational verb.

2.3 Multi-tenancy, ownership, and exit

Multi-tenant architecture matured as an economics of shared infrastructure [5], with isolation as its safety story; data-protection law made ownership-like rights enforceable [10]. Two older literatures supply what software multi-tenancy lacks. Ostrom’s institutional analysis of common-pool resources [27] shows that commons survive through graduated sanctions, monitoring, and boundary rules — machinery echoed in the membrane’s meters and valves. Hirschman’s exit-voice duality [16] underwrites the thesis’s central sovereignty device: a verified, low-cost *exit* (the secede port) is what keeps *voice* (the charter council) honest.

2.4 Federated learning and privacy machinery

Cross-tenant learning without raw-data exchange is the federated-learning premise [25], hardened by secure aggregation [4] and made quantifiable by differential privacy [7, 8], whose ϵ -budget is adopted directly as the membrane's leakage meter. Fair attribution of joint gains draws on the Shapley value [31] and its data-valuation instantiation [13], adopted for the contribution accounts. These mechanisms are mature in isolation; their composition under unilateral revocability and incubation quarantine is the membrane's contribution.

2.5 Markets and prices for shared computation

Pricing computation is an old idea — a futures market in machine time was proposed in 1968 [33] — and market-based resource allocation matured in systems such as Spawn [34]. Shadow prices as decentralized admission signals have rigorous foundations in network rate control [17]. The thesis adopts this tradition as a single posted price $\lambda(t)$ rationing compute and recall alike, with one deliberate deviation from pure market logic: quality floors and deadlines are lexically prior to price (Chapter 8 examines the market-first alternative and rejects it on exactly this point).

2.6 Evolving systems, drift, and governed change

That successful software must evolve continually is Lehman's law [20]; that deployed learning systems drift is equally established [12]. Singular learning theory [35] and its recent estimators [19] suggest that capability change is punctuated and measurable — adopted here as the drift gauge that arms quarantine. Goodhart effects [32, 24] motivate the adversarial gauntlet standing between measurement and law. Provenance modelling [26], digital twins [14], immune-inspired self-nonself discrimination [11], coevolutionary theory [30], and classical feedback analysis [1] supply, respectively, the audit substrate, the trial arena, the incubation posture, the vocabulary of coupled adaptation, and the loop-gain criterion the resonance damper enforces. Documented design rationale [28] motivates the fossil record.

2.7 The gap

Every mechanism above exists; no architecture composes them under the ten obligations of Chapter 3 simultaneously. Federated learning assumes a coordinator the sovereignty obligation forbids; market scheduling prices what the time-boundedness obligation says must be guaranteed; multi-tenant isolation prevents what the coupling obligation demands; and no surveyed system makes its own design evolution a governed function of its operators' recorded experience. The gap is a synthesis gap. Filling it is the work of Chapters 3 through 9.

3 · Requirements Analysis: The Ten-Point Decomposition

The brief is decomposed into ten load-bearing obligations, each with the design question it opens and a traceability map to the architecture that discharges it.

3.1 Reading the brief as an engineering document

Read as an engineering brief, the originating sentence of Section 1.2 contains exactly ten load-bearing commitments. Each is stated below as an obligation together with the design question it opens; the architecture of Chapters 4–7 is, clause by clause, the discharge of these obligations.

3.2 The ten obligations

P1 · Maturity. The system is specified at governed steady state, not as a prototype that will acquire governance later. Rollback, audit, pricing, and exit are constitutive, present from the first day of operation. *Design question:* what must be true on day one for year-ten operation to be lawful?

P2 · End-to-endness. The framework closes its own loop: query admission, answer, deposited experience, and design change form one circulation with no seam that crosses an unspecified boundary. *Design question:* where, precisely, does the loop close, and what carries it across each joint?

P3 · Co-adaptivity. Agents adapt to the system and the system adapts to the agents; adaptation is bidirectional by construction, not by accident [30]. *Design question:* through what medium, at what feedback gain, and under whose control?

P4 · Tenant multiplicity. More than one principal inhabits the framework as a first-class citizen; tenancy is a unit of ownership and law, not a namespace prefix. *Design question:* what is the unit of tenancy, and what does it contain?

P5 · Tenant coupling. Tenants are coupled: value flows between them. The brief says *coupled*, not *pooled* — exchange with structure, not fusion. *Design question:* what may cross between tenants, at what price, and what may never cross?

P6 · Client ownership. The client owns the search estate — corpus, indexes, learned models, experience records, and the right to leave with all of it [16]. *Design question:* what does ownership mean operationally: which keys, which export, which revocations?

P7 · Agent operation. Autonomous agents are the operators — they issue and refine queries, curate indexes, tune rankers, and audit outcomes. Humans hold authority but no operational verbs. *Design question:* what division of labour among agent roles, and what is irreducibly reserved to humans?

P8 · Time-boundedness. Every search carries a deadline contract, and every retained artifact carries a validity horizon. Time is a first-class term of service, not a performance aspiration. *Design question:* what happens — exactly — when the bound cannot be met?

P9 · Search as the organizing function. The workload is retrieval and foraging; the architecture is organized around what searching agents need rather than what human querying habits assume. *Design question:* what index geometry and memory stratification serve machine foraging?

P10 · Experience-shaped evolution. The design itself evolves, and the record of agent experience is the raw material — and the only legitimate warrant — for that evolution. *Design question:* how does lived trace become binding design without silent drift?

3.3 Traceability

Table 3.1 maps each obligation to its primary and secondary discharge sites; Figure 3.1 renders the same map graphically. The convention matters: an obligation with no named discharge site would be rhetoric, and the table is the check that none is.

Table 3.1: Obligations P1-P10 and their discharge sites in the architecture.

P	Obligation	Primary discharge	Secondary
P1	maturity	Invariant lattice I1-I7 (Sec. 7.4)	Charter plane (Sec. 5.1); rollback rail (EV-6)
P2	end-to-endness	The three circulations (Sec. 7.1)	Evolution pipeline (Sec. 6.2)
P3	co-adaptivity	Fast loop ES-7 + pipeline (Sec. 6.2)	Resonance damper EV-7 (I7)
P4	tenant multiplicity	Estates, Plane 1 (Sec. 5.2)	—
P5	tenant coupling	Membrane, Plane 2 (Sec. 5.3)	Contribution accounts MB-5
P6	client ownership	Estate boundary ES-0; secede port ES-6	Invariant I1; revocation MB-6
P7	agent operation	Agent crew ES-5 (Sec. 5.2)	Charter exclusions (Sec. 5.1)
P8	time-boundedness	Temporal spine (Sec. 6.1)	Anytime kernels SB-1; invariant I2
P9	search as function	Stratified index ES-2 (Sec. 5.2)	Kernels SB-1; rung-stratum binding
P10	experience-shaped evolution	Evolution pipeline (Sec. 6.2)	Experience ledger ES-4; invariant I5

The ten points of the original conceptualization, and where each is discharged

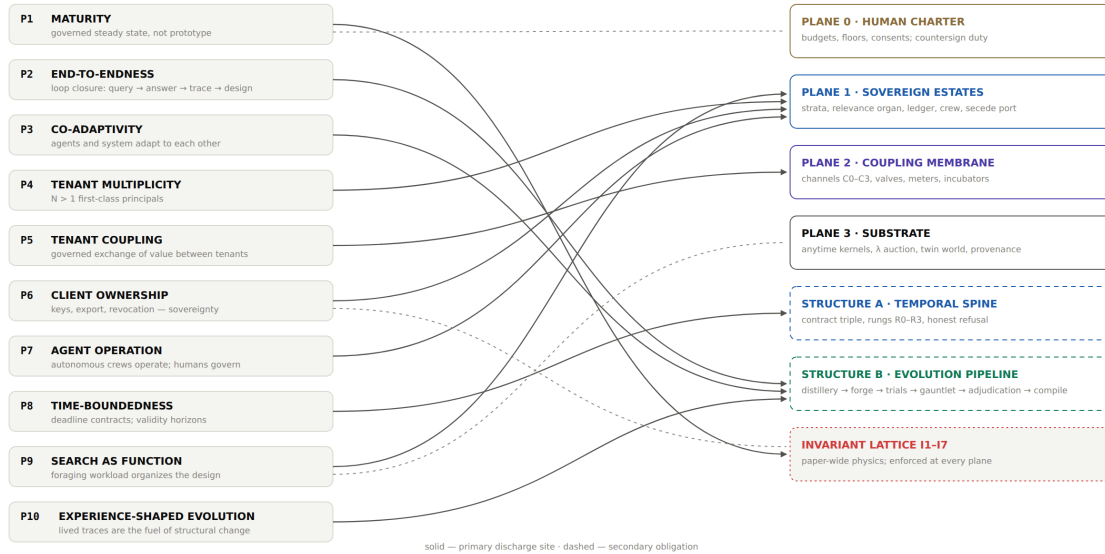


Figure 3.1: The ten obligations and their discharge sites. Solid lines mark the primary site where an obligation is discharged; dashed lines mark secondary obligations. Every point lands on a named plane, transverse structure, or the invariant lattice — none is discharged rhetorically.

4 · Architectural Overview

One governing rule, four planes, two transverse structures, three circulations. This chapter fixes the shape; Chapters 5–7 fill it in.

4.1 The governing rule

One rule governs the whole synthesis, and every later section can be derived from taking it seriously:

Share one admission calculus everywhere — value per unit cost against a posted price — but never confuse throughput with authority. Authority over the system’s future is granted only by verification, and the burden of verification grows with a change’s blast radius, not with its frequency.

The rule produces a deliberate asymmetry. Millions of queries per day are admitted by a cheap local test; a single design mutation per month is admitted by replayed trials, adversarial refutation, and countersigned adjudication. Between those extremes sit model updates (quarantined) and cross-tenant exchanges (metered).

4.2 Planes, structures, circulations

The architecture that carries this asymmetry consists of four horizontal planes — human charter, sovereign estates, coupling membrane, operator substrate — crossed by two transverse structures: a temporal contract spine through which all queries descend, and an evolution pipeline through which all design ascends. Three circulations tie them together: queries fall (blue), experience rises (amber), design returns (green). Figure 4.1 draws the whole.

A note on the frame drawn in Figure 4.1: the estates, the membrane, and their prices sit **inside** the generation frame $G(n)$ because they are products of design and therefore evolve; the human charter and the operator substrate sit **outside** it because they are parties to the design, providing respectively its legitimacy and its physics. This placement is not decorative — it states which parts of the world the evolution pipeline is permitted to rewrite.

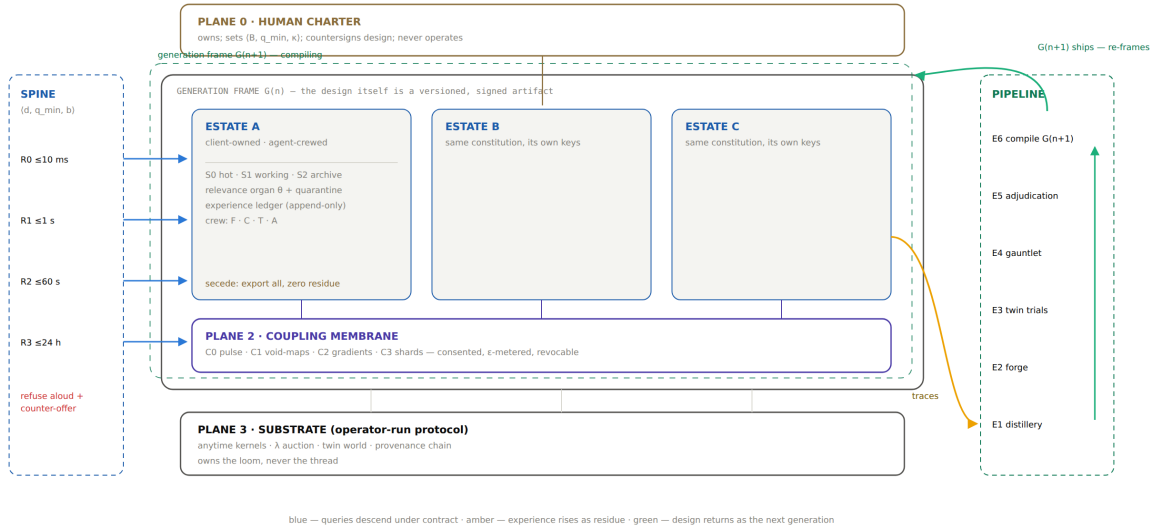
Macro-architecture: four planes, two transverse structures, three circulations

Figure 4.1: Macro-architecture. Four planes and two transverse structures, wrapped in a versioned generation frame: the design itself is a signed artifact, and the next frame is always compiling. The three circulations — query, experience, design — are specified edge-by-edge in Chapter 7.

4.3 Specification conventions

Components are numbered within their plane (ES-2, MB-4, SB-3, SP-1, EV-6). Chapters 5 and 6 state each plane's mandate, introduce every component and subcomponent with its function, and close with the plane's exclusions — what it may never do. No cross-plane relationship is asserted there beyond naming; all edges are defined once, in Chapter 7, whose registry is closed-world and normative.

5 · The Planes

Each plane in dedicated treatment: mandate, components and subcomponents with their functions, and exclusions.

5.1 Plane 0 — The Human Charter Plane

Mandate. Plane 0 is where ownership and accountability live. Its occupants are people — the clients who own estates and the officers they appoint. The plane’s entire output is a small set of signed instruments; it performs no operations.

CH-1 · Charter Council. The standing body of client principals (one seat per tenant, plus appointed independent verifier seats). Function: to hold the signing keys of governance — approving charters, countersigning design generations, and seating disputes. Subcomponents: *seats* (identity-bound, non-delegable to agents), *quorum rules* (charter-defined, recorded), and the *countersign duty*: no design generation ships without council signature (see EV-5, Section 6.2).

CH-2 · Charter Instrument. The signed document each tenant issues; the only channel through which human intent enters the system. Function: to set the three operating quantities — the budget envelope **B** (what the tenant will spend), the quality floors **q_min** per query class (below which an answer is a breach, not a bargain), and the coupling consents **k** (which membrane channels are open, toward whom, at what privacy budget) — together with a liability posture for refusals and breaches.

CH-3 · Dissent Register. An append-only record of council votes and minority positions. Function: to make disagreement durable — dissent is logged, never overwritten, so that a later failure can be traced to the argument that predicted it.

Exclusions. Plane 0 has no operational verbs: it cannot issue queries, touch strata, read another tenant’s matter, or tune a model. Humans govern; they do not operate. This is the discharge of P7’s reserved-to-humans clause: legitimacy, not labour.

5.2 Plane 1 — The Sovereign Tenant Estates

Mandate. Plane 1 is where all tenant matter lives and all searching happens. Each tenant occupies one estate: a cryptographically bounded territory containing corpus, indexes, models, experience, and the agent crew that works them. Estates are identical in constitution and disjoint in content — the same law, never the same matter. This plane discharges P4 (multiplicity), P6 (ownership), P7 (agent operation), and P9 (search-as-function).

ES-0 · Boundary and keys. The estate’s perimeter. Function: everything inside is tenant property under tenant keys K_T ; every artifact that leaves is signed, every artifact that enters is quarantined (see MB-4). The boundary is what makes the word “owned” in the brief operational rather than merely contractual.

ES-1 · Corpus Vault. The client’s raw matter — documents, records, media — encrypted under client keys. Function: source of truth; nothing is searchable until a curation verb promotes it into the strata, so the vault can hold material the tenant never exposes to search at all.

ES-2 · Stratified Index (S0/S1/S2). The estate’s searchable memory, stratified by residence time rather than by data type. **S0, the hot lattice** ($\tau \approx ms$): sketches, fingerprints, and caches able to answer in microseconds; its epistemic standard is sanity, not scholarship. **S1, the working index** ($\tau \approx s$): approximate-nearest-neighbor graphs [23] and posting structures built for early-exit descent, where answer quality improves monotonically with time spent. **S2, the deep archive** ($\tau \approx min-d$): exhaustive, cold, reachable only by campaign-class queries. Function of the stratification: depth is purchased by deadline, never presumed (the spine, Section 6.1, enforces this); and each stratum carries its own validity predicate, so “correct” at ten milliseconds is a different claim than “correct” overnight. *Adopt, not invent*: anytime retrieval structures are established [6, 38]; their stratification under deadline law is the contribution.

ES-3 · Relevance Organ 0. The estate’s learned ranking model, client-signed. Subcomponents: **ES-3a, the drift gauge** — an online estimator of learning capacity (a λ -hat statistic in the spirit of learning-coefficient estimation [35, 19]; *adopt, not invent*) whose spikes signal that the model is undergoing a capability transition rather than a smooth improvement; and **ES-3b, the quarantine cell** — a holding stage through which every parameter delta must pass, releasing updates only between missions and only when the gauge is quiet (invariant I6). Function: relevance learning at full speed with structurally bounded surprise.

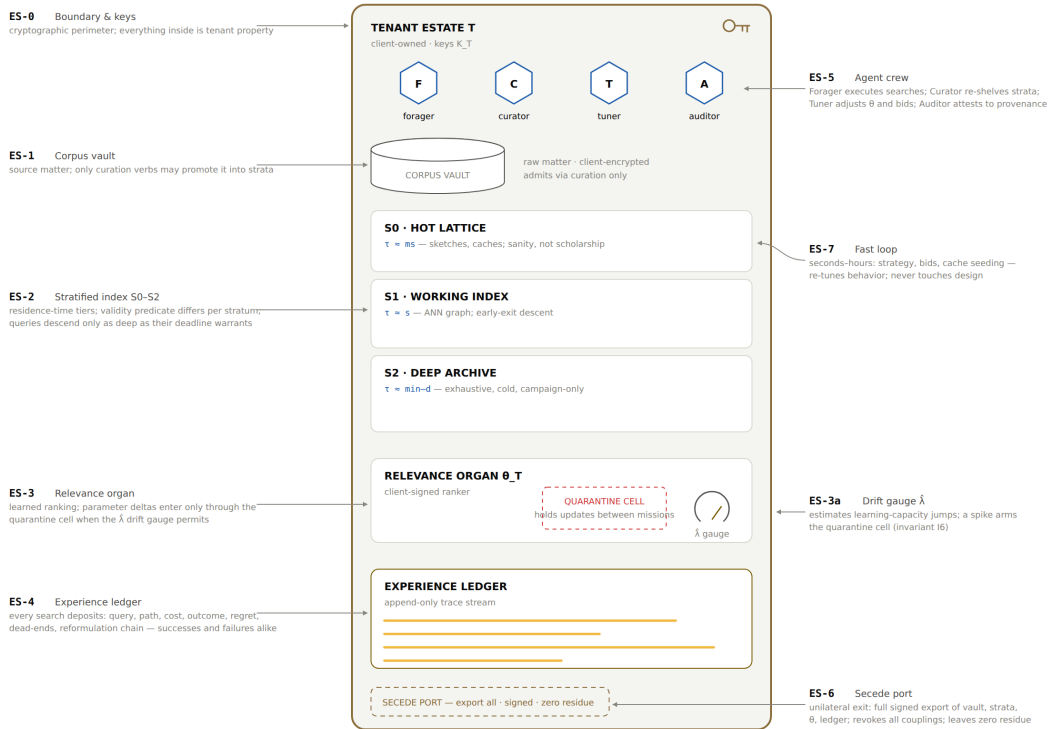
ES-4 · Experience Ledger. The estate’s append-only memory of its own operation, and the architecture’s single most consequential component: it is the sole legitimate fuel for design evolution (P10). Every search deposits a trace with

six mandatory fields: the query contract as admitted, the descent path taken, the cost actually spent, the outcome delivered (including refusals), the regret estimate (what better answer later surfaced, if any), and the dead-ends and reformulation chain en route. Failures are first-class deposits: a dead-end atlas is an asset, not an embarrassment (invariant I4). A tail-event annex pins rare, expensive failures so they survive later compression.

ES-5 · Agent Crew. The estate’s operators — four charters, any number of instances [36]. **The Forager** executes searches: it descends strata under contract, assembles anytime answers, and owns the deposit duty. **The Curator** works the strata: promotion and demotion between $S_0/S_1/S_2$, cache seeding, vault-to-stratum admission, eviction hygiene. **The Tuner** works the organ and the wallet: it shapes θ within quarantine law and bids the estate’s budget at the spine’s gates. **The Auditor** attests: it verifies deposits, signs exports, and is the estate’s witness to the provenance chain. Clients may bring their own crew implementations; the charters, not the binaries, are the law.

ES-6 · Secede Port. The exit [16]. Function: on unilateral demand, produce a complete signed export — vault, strata, θ , ledger, provenance extracts — revoke all couplings, and leave zero residue behind (invariant I1). The port is load-bearing precisely because it is never supposed to be used: its verified existence is what makes every other promise credible.

ES-7 · The Fast Loop. Not a part but a licensed behaviour: on timescales of seconds to hours, the crew retunes strategy — query plans, bids, cache policy, reformulation habits. Function: give co-adaptation its rapid, cheap half. The fast loop touches strategy only; by construction it cannot alter schema, strata geometry, API verbs, prices, or anything else that belongs to the design (that power lives only in the evolution pipeline, Section 6.2).

Plane 1 — anatomy of a sovereign tenant estate (components ES-0 ... ES-7)**Figure 5.1:** Anatomy of one estate. Components ES-0 through ES-7; callouts state each part’s function. All estates share this constitution; none shares its content.

Exclusions. An estate may not read another estate’s matter (all exchange is membrane-mediated), may not mutate its own design (fast loop is strategy-only), and may not silently discard experience (the deposit duty is unconditional).

5.3 Plane 2 — The Coupling Membrane

Mandate. Plane 2 is the entire answer to the brief’s hardest phrase, “multi-coupled-tenant”: tenants must trade value without trading sovereignty. The membrane is a graded, consented, metered, revocable exchange surface between estates — coupled, never merged. It discharges P5 and, jointly with ES-0/ES-6, protects P6. Its institutional logic — boundary rules, monitoring, graduated commitment — follows the design principles long observed in durable common-pool institutions [27].

MB-1 · The four channels. Coupling is not one thing but a ladder of four, ordered by risk and by the consent strength required to open them. **C0 · Pulse** — differentially private telemetry aggregates (load, latency distributions, coarse hit rates) [7, 8]. Always on; the federation’s weather report; lowest risk. **C1 · Void-maps** — negative knowledge: dead-end atlases, exhausted-region markers,

futile-reformulation warnings. Opt-in; remarkably high value per bit, because knowing where nothing lives rarely identifies what does. **C2 · Gradient federation** — signed model deltas exchanged under secure aggregation [25, 4], letting relevance organs learn jointly without exposing traces. **C3 · Consortium shards** — leased, jointly searchable sub-indexes; the strongest coupling short of merger, under the strongest consent. Channel width in Figure 5.2 encodes this ladder.

MB-2 · Valves and consent keys κ . Every channel between every tenant pair carries a valve that opens only under a live, charter-issued consent key κ_{AB} . Function: make coupling an explicit, inspectable state rather than an emergent property of shared infrastructure.

MB-3 · ϵ -Meters. Privacy-budget accounting on every open valve: each exchange spends from a finite ϵ allowance set in the charter; an exhausted meter closes the valve until the council renews it. Function: leakage is bounded by arithmetic, not by hope.

MB-4 · Incubators. Quarantine chambers on the receiving side of every channel, in the immune posture of self-nonsel self discrimination [11]. Foreign signals — a void-map, a gradient delta, a shard update — sit in incubation while the estate’s auditor checks provenance and the drift-gauge baseline. Function: nothing foreign touches an organ or a stratum directly.

MB-5 · Contribution Accounts. Shapley-style give-and-take accounting per tenant pair and per channel [31, 13]. Function: keep coupling reciprocal over time and make free-riding legible before it becomes resentment — the political failure mode of federations.

MB-6 · Revocation Lever. Unilateral, immediate, logged closure of any consent by its issuing tenant. On revocation the channel drains: in-flight exchanges complete or abort per charter, incubators flush, and the ϵ -meter freezes for audit. Function: consent that cannot be withdrawn instantly is not consent.

Plane 2 — the coupling membrane between any two estates (components MB-1 ... MB-6)

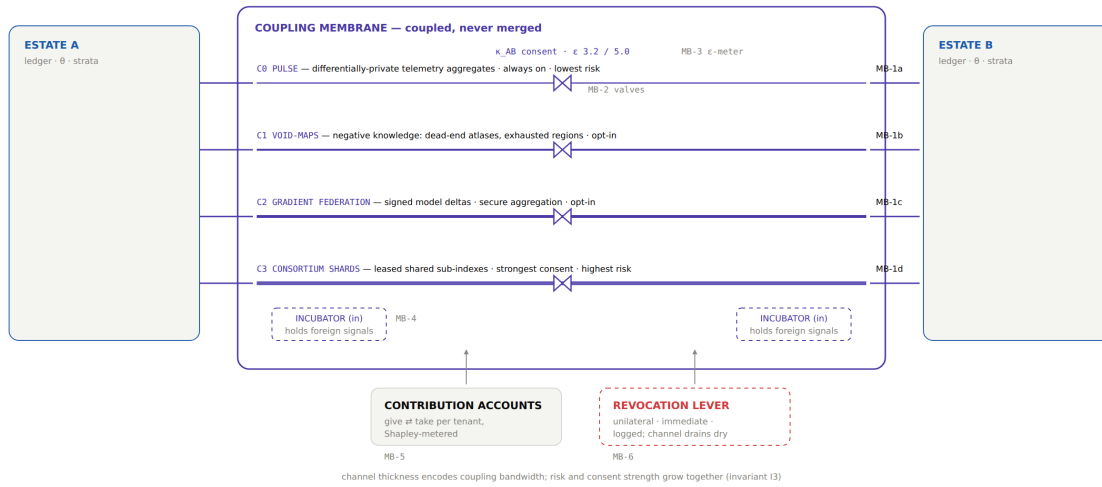


Figure 5.2: The membrane between any two estates. Four channels in rising order of coupling bandwidth and risk; valves under consent keys; ϵ -meters bounding leakage; incubators quarantining inbound signals; contribution accounts and the revocation lever.

Exclusions. The membrane carries derived signals only — aggregates, negative knowledge, deltas, leases. Raw corpus matter never crosses it on any channel; a tenant wanting another’s documents is asking for a contract, not a coupling.

5.4 Plane 3 — The Operator Substrate

Mandate. Plane 3 is what the framework operator runs: the shared physics under all estates. Its one-sentence constitution: *the operator owns the loom, never the thread* — every substrate surface is protocol, and no substrate surface is tenant data.

SB-1 · Anytime Kernels. The library of interruptible algorithms every estate builds on: early-exit ANN descent, sketch-then-sweep ladders, progressive scoring [6, 38, 23]. Function: guarantee that any computation can stop at deadline d and hand back the best answer so far, with quality monotone in time spent. This is the mechanical premise of the entire spine (Section 6.1).

SB-2 · The λ Auction. The scheduler, expressed as one shadow price [17, 33, 34]. Every claim on shared capacity — compute, recall, bandwidth — is admitted by the same local test: $\text{admit} \iff \Delta \text{value} / \text{cost} \geq \lambda(t)$. The price rises under load and falls in slack; tuners bid estate budgets against it. Function: ration everything with one lever; make “the system is busy” a number rather than a mood. Estimation asymmetry is law: when uncertain, understate value and overstate cost, so ambiguity starves rather than floods the system.

SB-3 · The Twin World. A shadow replica of the federation [14] — synthetic estates, recorded workloads, membrane simulacra — in which candidate designs are exercised. Function: the only arena where the evolution pipeline’s trials (EV-3) run; nothing tried in the twin has live effect. The twin consumes traces under the same consents κ that govern the membrane.

SB-4 · The Provenance Chain. A hash-linked, third-party-auditable record of every consequential event [26]: admissions, evictions, refusals, consents, revocations, quarantine releases, rollouts, secessions. Function: the paper trail that converts every promise in this document from assurance into evidence.

SB-5 · The Loom Protocol. The operator’s complete external surface: the wire formats, contract grammars, and proof obligations that estates and membrane implementations must satisfy. Function: substitutability — any module (including the operator’s own) is replaceable by any implementation that honors the protocol and its proofs, which is what keeps the operator disciplined.

Plane 3 — the operator substrate (components SB-1 ... SB-5)

SB-5 · THE LOOM PROTOCOL — the operator’s entire surface; tenant matter never crosses this line

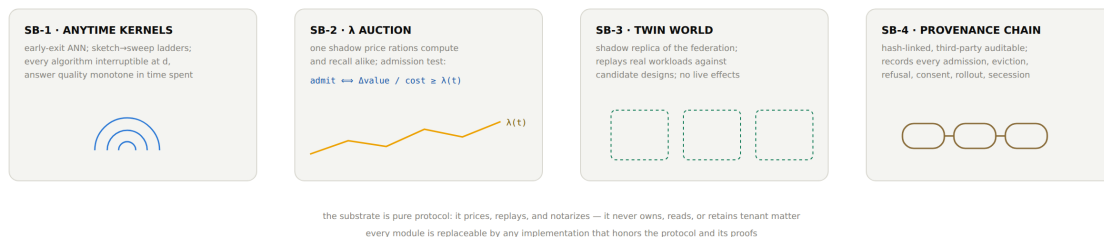


Figure 5.3: The substrate. Four modules beneath a single boundary rule: everything above the dashed line is tenant matter and never crosses it; everything below is protocol.

Exclusions. The substrate never owns, reads, retains, or prices tenant content as content; it prices claims on capacity. It cannot originate design proposals (it has no experience of its own) and holds no seat on the council.

6 · The Transverse Structures

Two structures cross all planes: the temporal contract spine, down which every query descends, and the evolution pipeline, up which all design ascends.

6.1 Structure A — The Temporal Contract Spine

Mandate. The spine is time-boundedness (P8) made structural: a single admission-and-descent path through which every query enters the fabric, carrying an explicit contract, and down which it may purchase depth. It is drawn vertical because that is what it is: a ladder of deadlines aligned to the strata of every estate, in the tradition of hard-real-time admission [22].

SP-1 · Query Contract. Every query arrives as the triple $\langle d, q_{\min}, b \rangle$ — deadline, quality floor, budget. Function: convert the vague “fast and good” into three numbers the rest of the machine can enforce. Contracts are written by requesting agents (or standing charter defaults) and are binding on both sides: the fabric may not exceed d or undercut $q_{\min}$; the requester may not consume more than b .

SP-2 · Rungs R0-R3. Four deadline classes, each admitting specific strata and kernels. **R0 Reflex** (≤ 10 ms): S0 only — sketch probes and cache hits; the contract is sketch-or-silence, and no speculative depth is permitted. **R1 Interactive** (≤ 1 s): S0+S1 under early-exit descent; the answer in hand improves monotonically until the deadline expires. **R2 Deliberative** (≤ 60 s): bounded S2 access, multi-hop foraging, reformulation chains within budget. **R3 Campaign** (≤ 24 h): full archive sweeps and — where consents permit — coupled recall across the membrane. Function: make “how deep may this query go” a consequence of its contract, not of its enthusiasm.

SP-3 · Admission Gates. At every rung, one local test: $\text{ADMIT} \iff \text{bid} \geq \lambda(t) \wedge E[q \mid d, b] \geq q_{\min}$ — the bid clears the substrate’s price and the expected quality (estimated from ledger history) clears the floor. Estimation errors round conservative. Function: keep both scarcity and honesty at the door.

SP-4 · Honest Refusal. When the gate’s quality test fails, the spine refuses aloud and returns a counter-offer $\langle d', q' \rangle$: the nearest contract it believes it can honor. The system may degrade to the floor or decline explicitly; it may never be silently late or silently shallow (invariant I2). Every refusal is a logged, first-class outcome — and a ledger trace, because refusals are experience too. Function: preserve trust under load; a fabric that lies about time will be routed around, then

abandoned.

SP-5 · Result Stamps. Every answer leaves stamped (TTL, confidence, cost, provenance ref). Function: validity is explicit and expiring, never assumed — the consuming agent knows exactly how long the answer is warranted and what it cost to make.

Structure A — the temporal contract spine (components SP-1 ... SP-5)

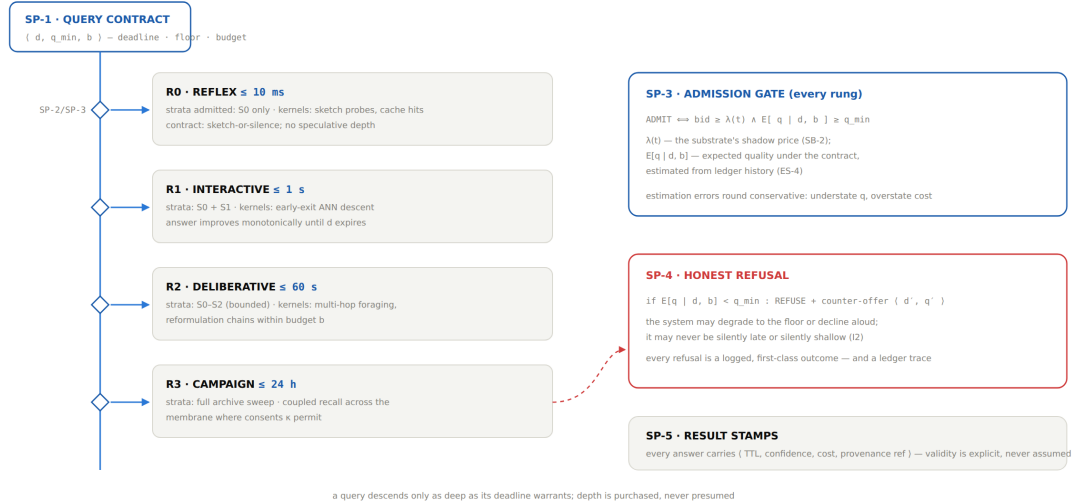


Figure 6.1: The temporal contract spine. A query's contract admits it to a rung; each rung admits specific strata and kernels; the gate tests price and expected quality; failure produces an honest refusal with a counter-offer, never silence.

6.2 Structure B — The Evolution Pipeline

Mandate. Structure B is P10 made governable: the six-stage path by which recorded experience — and nothing else — becomes binding design. It is the slow loop, deliberately weeks-per-cycle where the fast loop is seconds; the separation of timescales is load-bearing, because co-adaptation without timescale separation is resonance [1]. That the design must keep evolving at all is not optional but lawlike [20]; the pipeline exists so that the evolution is governed.

EV-1 · Distillery. Compresses ledger streams (scoped by consent) into decision-grade forms: regret spectra by query class, friction maps over strata and API verbs, refusal and dead-end statistics. Maintains the tail-event vault so rare failures survive compression. Function: turn millions of traces into a shape a proposal can cite.

EV-2 · Proposal Forge. Drafts design mutations: strata geometry, schema fields, API verbs, valve types, price schedules, rung boundaries. The forge's law is invariant I5's first clause: *no proposal without cited traces* — every mutation

names the distillate evidence that motivates it. Function: keep design change answerable to lived experience rather than to fashion.

EV-3 · Twin Trials. Candidate designs replay recorded workloads in the twin world (SB-3), including counterfactual load and adversarial traffic mixes. Function: measure a proposal’s consequences before any tenant feels them.

EV-4 · Gauntlet. Independent adversarial verifiers attempt refutation: leakage across κ boundaries, regression against quality floors, capture of the proposal by the metric it cites [24, 32]. Refuters default to “refuted” when uncertain. Function: institutionalized disbelief between measurement and law.

EV-5 · Adjudication. The charter council (Plane 0) countersigns surviving proposals under its quorum rules; dissent enters the register. Function: the constitutional joint — the one point where human authority touches design, and the discharge of P1’s legitimacy clause.

EV-6 · Generation Compiler. Compiles the countersigned change-set into the next generation frame $G(n+1)$: staged rollout, rollback rail armed, signed release. Function: design change as an atomic, reversible, attributable event — never a drift.

EV-7 · Resonance Damper. The governor on the whole double loop: release requires the estimated co-adaptive loop gain — how strongly design responds to experience times how strongly experience responds to design — to stay below one (invariant I7) [1]. Function: co-adaptation is the point; runaway co-adaptation is the failure mode. The damper is what lets the fabric court the first without suffering the second.

EV-8 · Fossil Record. Refuted proposals and retired organs, kept legible with their evidence and dissents — the documented rationale a rational-looking design process requires [28]. Function: institutional memory; a rejected idea returns only against the argument that killed it.

EV-9 · Generation Frames. The versioned design artifacts themselves: $G(n)$ live and signed, $G(n+1)$ compiling, predecessors retired to fossils. Function: make “the design” a noun with a hash, so that P10’s “evolving design” is a sequence of auditable objects rather than a mood.

Structure 8 — the experience–design evolution pipeline (components EV-1 ... EV-9)

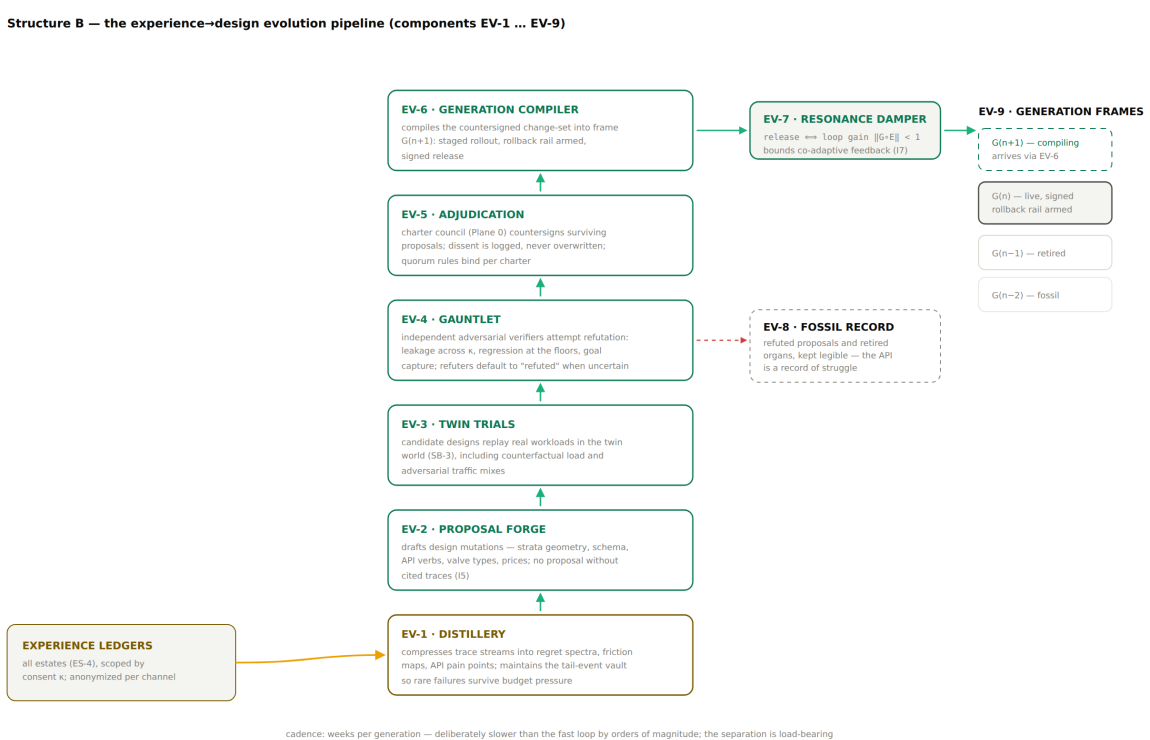


Figure 6.2: The evolution pipeline. Experience ascends through distillation, proposal, trial, refutation, and adjudication before compiling into a signed generation; the damper bounds the loop gain; fossils keep every refusal legible.

7 · Relations and Invariants

All relationships among components, stated once under a closed-world convention, and the seven invariants that bind every plane.

7.1 The closed-world convention and the three circulations

Components having been introduced, their relationships can now be stated under a closed-world convention: **an interface exists if and only if it appears in Table 7.1**. Absence is prohibition. Three composite circulations give the registry its narrative shape. The **query circulation** descends: contract → gate → strata → stamped answer (IF-04, -05, -06, -26). The **experience circulation** ascends: deposit → ledger → distillery (IF-07, -08, -18). The **design circulation** returns: distillate → proposal → trial → gauntlet → adjudication → compiled generation → every plane (IF-19 through IF-24). Everything else is governance and sovereignty plumbing: charter instruments flowing down, consents and revocations flowing sideways, provenance flowing always.

Principal interfaces — nodes are components, numbered edges key to the Interface Registry (Table 7.1)

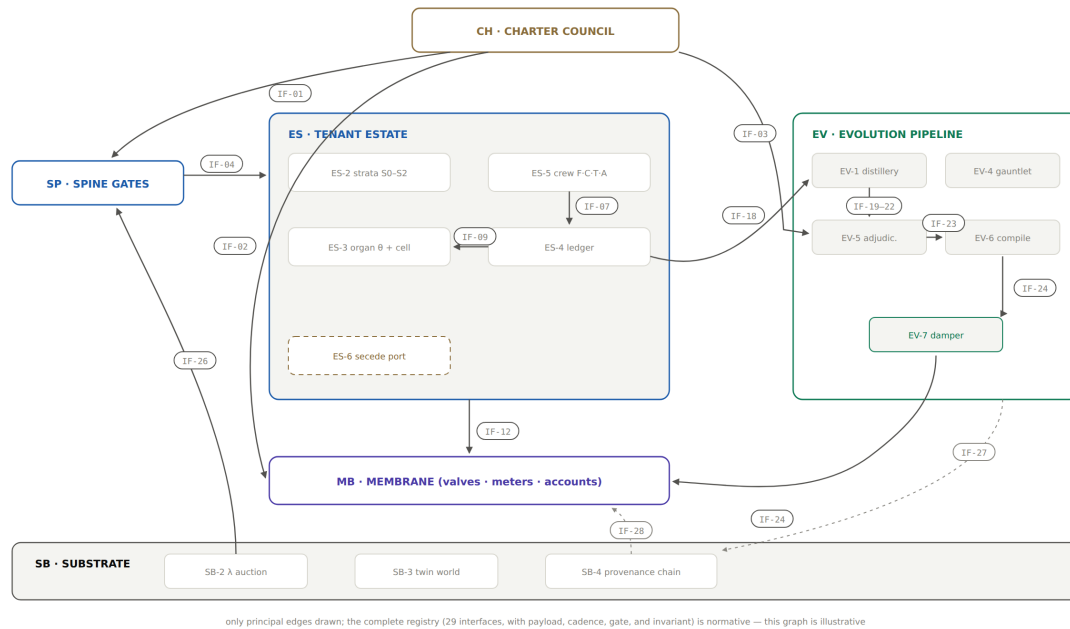


Figure 7.1: Principal interfaces. Nodes are components from Chapters 5–6; numbered edges key into Table 7.1. The drawing shows the principal edges only; the table is normative and complete.

7.2 The Interface Registry

Table 7.1: The Interface Registry — normative and closed-world. An edge absent from this table does not exist and must not be implemented.

ID	Source → Sink	Payload	Cadence	Gate / Price	Inv.
IF-01	CH-2 → SP-3	floors $q_{\min}$, budgets B	on charter change	countersigned	I2
IF-02	CH-2 → MB-2	consents κ (open/scope)	on charter change	countersigned	I3
IF-03	CH-1 → EV-5	countersignature / dissent	per proposal	quorum	I5
IF-04	SP-3 → ES-2	admitted query (contract)	per query	$\text{bid} \geq \lambda \wedge E[q] \geq q_{\min}$	I2
IF-05	SP-4 → requester	refusal + counter-offer (d', q')	on gate failure	logged	I2
IF-06	ES-2/ES-5 → requester	stamped answer (TTL, conf, cost, prov)	per query	stamp mandatory	I2
IF-07	ES-5 → ES-4	trace deposit (6 fields)	per search, incl. refusals	unconditional	I4
IF-08	ES-4 → ES-5 (Tuner)	regret batches	fast loop (s-h)	read-only	I4
IF-09	ES-5 (Tuner) → ES-3	θ parameter delta	between missions	via ES-3b; gauge quiet	I6
IF-10	ES-5 (Curator) → ES-2	promotion / demotion / eviction	continuous	curation verbs only	I4
IF-11	ES-5 (Auditor) → SB-4	attestations	continuous	signed	I5
IF-12	ES-4 → MB-1a (C0)	DP telemetry aggregates	always on	ϵ -metered	I3
IF-13	ES-x $\nrightarrow$ MB-1b (C1)	void-maps (negative knowledge)	per exchange	κ + incubation	I3, I6
IF-14	ES-3 $\nrightarrow$ MB-1c (C2)	signed model deltas	per epoch	secure agg + incubation	I3, I6
IF-15	ES-2 $\nrightarrow$ MB-1d (C3)	shard leases	per lease	strongest κ	I1, I3
IF-16	MB-1 → MB-5	give/take entries	per exchange	Shapley metering	I3
IF-17	tenant → MB-6	revocation of κ	anytime	unilateral, immediate	I1, I3
IF-18	ES-4 (all) → EV-1	trace batches	slow loop (d-wk)	scoped by κ ; anonymized	I4, I5
IF-19	EV-1 → EV-2	distillates (regret spectra ...)	per cycle	citable forms only	I5
IF-20	EV-2 → EV-3	proposal bundle + cited traces	per proposal	twin-only execution	I5
IF-21	EV-3 → EV-4	trial results	per proposal	adversarial handoff	I5
IF-22	EV-4 → EV-5	surviving proposals + refutation log	per proposal	default-refuted	I5
IF-23	EV-5 → EV-6	countersigned change-set	per generation	quorum met	I5
IF-24	EV-6 → all planes	generation $G(n+1)$	staged rollout	damper gain < 1 ; rollback armed	I5, I7
IF-25	EV-4 → EV-8	refuted proposals + evidence	per refutation	archived legible	I5
IF-26	SB-2 → SP-3	shadow price $\lambda(t)$	continuous	none — it is the price	I2
IF-27	EV-3 $\nrightarrow$ SB-3	replayed workloads	per trial	sandboxed; consented traces	I5

ID	Source → Sink	Payload	Cadence	Gate / Price	Inv.
IF-28	all components → SB-4	provenance appends	always	hash-linked	I1, I5
IF-29	ES-6 → departing tenant	full signed export; couplings revoked	on demand	zero residue verified	I1

7.3 Reading the registry

Three properties deliver the “minimal ambiguity” the design owes its reader.

Completeness: every arrow in every figure corresponds to a row here; a relationship not listed does not exist and must not be implemented.

Directionality: payloads move only as written; the membrane rows (IF-12 through IF-15) are the only bidirectional edges in the fabric, and each is gated on both sides.

Guarded authority: the two edges that can change what the system *is* — IF-09 into a live model and IF-24 into a live design — are exactly the two carrying quarantine and damper conditions. Throughput edges are many and cheap; authority edges are few and expensive. That asymmetry is the governing rule of Section 4.1, now visible as table structure.

7.4 The invariant lattice I1-I7

Seven invariants bind every plane; each names its enforcing components. **I1 · Sovereignty.** Every byte of tenant state is exportable and client-signed; secession leaves zero residue. *Enforced by* ES-0, ES-6, MB-6, SB-4 (IF-17, -28, -29). **I2 · Honest time.** Meet the deadline, degrade to the floor, or refuse aloud with a counter-offer; never silently late, never silently shallow. *Enforced by* SP-1...SP-5, SB-1, SB-2 (IF-04, -05, -06, -26). **I3 · Graded coupling.** Coupling bandwidth is earned, consented, ϵ -metered, and unilaterally revocable; it widens only with demonstrated benefit and bounded leakage. *Enforced by* MB-1...MB-6 (IF-12...-17). **I4 · Experience is capital.** Every search deposits a complete trace; dead-ends and refusals are assets; the deposit duty is unconditional. *Enforced by* ES-4, ES-5 (IF-07, -08, -10, -18). **I5 · Design provenance.** No design mutation without cited traces, a twin trial, an adversarial gauntlet, and countersigned adjudication. *Enforced by* EV-1...EV-6, EV-8, CH-1, SB-4 (IF-19...-25, -28). **I6 · Drift quarantine.** Model-capability jumps are held between missions until verified; foreign signals incubate before absorption. *Enforced by* ES-3a, ES-3b, MB-4 (IF-09, -13, -14). **I7 · Bounded resonance.** The co-adaptive loop wears a damper: estimated agent $\rightleftharpoons$ design feedback gain stays below one at every release. *Enforced by* EV-7 (IF-24).

7.5 Order within the lattice

The lattice has an internal order that Chapter 8 will exploit: I1 and I2 are lexical — no quantity of benefit purchases their violation; I3-I6 are protective — they bound the mechanisms that make the fabric profitable; I7 is reflexive — it protects the system from its own central virtue.

8 · Design-Space Analysis: Alternative Syntheses of the Ten Points

The ten obligations admit many architectures. This chapter enumerates the stances each obligation allows, counts the combinations, prunes the incoherent, quotients the remainder into archetype families, and states the selection rule under which the Fabric — and not a rival — falls out.

8.1 Method

A specification that cannot say what else it could have been is advocacy, not architecture [28, 3]. This chapter treats the ten obligations of Chapter 3 as free variables, enumerates the architectural stances each admits, counts the raw combinations, prunes the incompatible, and collapses the remainder into archetype families — then states the selection rule under which the proposed architecture is the optimum. All counts are illustrative arithmetic over a stated enumeration, not measurements; their role is to make “how many other ways?” a question with a defensible shape rather than a shrug.

8.2 The stance matrix

Table 8.1: Each obligation admits three or four defensible architectural stances (chosen stance starred). The enumeration is conservative — stances differing only in vocabulary are merged.

Obligation	Admissible stances (chosen ★)	k
P1 maturity	prototype-first · retrofit-governance · constitutional ★	3
P2 closure	open pipeline · dual-loop ★ · triple-loop (meta-governor)	3
P3 co-adaptivity	parametric-only · protocol-negotiated · structural, gated ★ · unconstrained-emergent	4
P4 tenancy unit	namespace · account-in-market · sovereign estate ★ · organ-of-organism	4
P5 coupling topology	broker-mediated · pairwise membrane ★ · hub consortium · full-mesh market	4
P6 ownership	custodial · escrowed · sovereign-with-export ★ · self-hosted-protocol	4
P7 operation	single monolithic agent · homogeneous swarm · differentiated crew ★ · open agent market	4
P8 time regime	soft SLA · deadline ladder ★ · continuous-price-only · per-query negotiation	4
P9 search substrate	index-centric strata ★ · model-internalized (RAG-absorbed) · graph-centric · hybrid	4
P10 evolution	vendor-offline redesign · continual online learning · generational, gated ★ · population-evolutionary	4

8.3 Raw count, compatibility pruning, equivalence quotient

The raw space is the product of the stance counts: $3 \cdot 3 \cdot 4^8 = 589,824$ distinct stance-vectors. Not all are coherent. Six compatibility rules do most of the pruning. **R-a:** unconstrained-emergent co-adaptation (P3) is inadmissible under a constitutional maturity posture (P1) — one cannot guarantee what one refuses to bound. **R-b:** organ-of-organism tenancy (P4) collapses the coupling axis (P5): organs do not trade, they share; three of four P5 stances vanish. **R-c:** custodial ownership (P6) removes the client’s standing to countersign, striking gated-generational evolution (P10) from its row. **R-d:** continuous-price-only time (P8) cannot express quality floors and is incompatible with dual-loop closure (P2), which needs floors to define regret. **R-e:** model-internalized search (P9) under sovereign-with-export ownership (P6) forces whole-model export rights, striking escrowed hybrids. **R-f:** open agent markets (P7) under constitutional maturity (P1) require a licensing layer that collapses them, in effect, into differentiated crews. Applied jointly, the rules eliminate roughly four of every five vectors, leaving on the order of 1.2×10^5 admissible configurations — still absurdly many, which is the point: stance-vectors are not designs.

The meaningful count comes from a quotient. Call two configurations equivalent when they differ only by plane-merging or plane-splitting, by renaming, or by dual representation of the same governance relation (a valve “in the membrane” versus a filter “at the estate wall” is one mechanism drawn twice). Under this equivalence the admissible space collapses into approximately ten families, of which seven are load-bearing — internally coherent, defensible end-to-end, and genuinely distinct in what they optimize and what they sacrifice. The honest summary of “how many different other ways”: **589,824 raw syntheses; about 1.2×10^5 jointly admissible; seven that matter.**

8.4 The seven archetype families

A · The Fabric (chosen; this thesis). Federal physiology: sovereign estates, graded membrane, deadline spine, gated generational evolution. Buys the forbidden corner — high coupling at high sovereignty — at the price of heavy governance machinery. Disqualifier it avoids: none within its own priorities; its risk is apparatus weight, examined in Section 10.3.

B · The Exchange. Market-first synthesis in the lineage of computational economies [33, 34]: every interaction — retrieval, coupling, even design change — is an auction; coupling is trade; tenancy is an account. Buys allocative efficiency and effortless prioritization under load. Forfeits floors: when everything is priced, quality floors and honest time become price-contingent, and a rich adversary can starve a poor tenant’s deadlines lawfully. Disqualified because I2 must be lexical,

not purchasable.

C · The Commons. Ecological synthesis: one shared mycelial index; ownership as cryptographic tagging and policy filters rather than territory [27]. Buys maximal learning velocity — every trace fertilizes every tenant. Forfeits hard sovereignty: export is possible but excision is not; a departing tenant’s influence remains diffused through shared structure forever. Disqualified on P6: exit without residue is unachievable by construction.

D · The Guild Blackboard. Classical multi-agent synthesis [9]: one shared blackboard; role-guilds negotiate posts and claims; design evolves by consensus editing. Buys transparency and architectural simplicity. Forfeits both isolation and time-boundedness at scale — negotiation latency is unbounded, and the blackboard is one tenancy in all but name. Disqualified on P4 and P8 jointly.

E · The Archipelago. Isolation-first synthesis: estates as islands; coupling only through episodic, human-approved embassy missions carrying vetted parcels. Buys near-absolute sovereignty and auditability. Forfeits co-adaptation: experience pools too slowly for the design loop to learn anything before the world changes. Disqualified on P3/P10 starvation — the brief demands a living loop, not an annual summit.

F · The Organism. Total integration in the autonomic lineage [18]: tenants as organs of one body — single ledger, single relevance organ, unified crew. Buys the strongest co-adaptation physically available; nothing couples like shared flesh. Forfeits ownership entirely: secession is surgery, with scars on both sides. Disqualified on P6’s exit test — the first tenant to leave would falsify the architecture [16].

G · The Protocol Mesh. Operator-less synthesis: the substrate dissolved into open standards; estates self-host; evolution by versioned RFC. Buys credible neutrality — no operator to capture. Forfeits enforcement: invariants become conventions, the twin world has no owner, and the verification-burden asymmetry of Section 4.1 has no enforcer. Disqualified on P1: a mature system must be able to make its own promises binding.

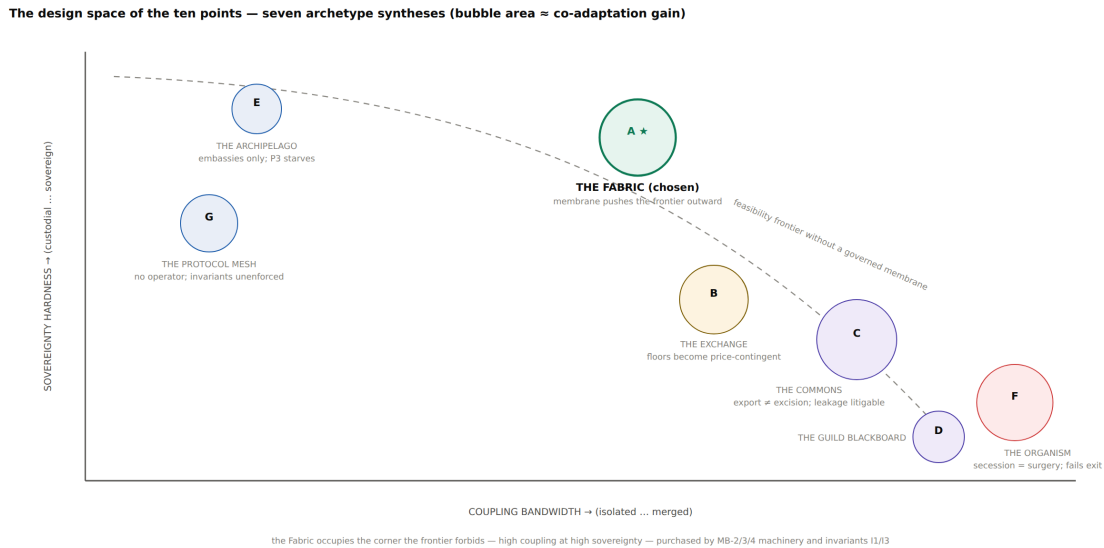


Figure 8.1: The seven archetypes on the two axes the brief cares most about. The dashed frontier is the trade-off available without governed-coupling machinery; the Fabric's wager is that membrane components MB-2/3/4 and invariants 11/13 push that frontier outward.

8.5 The selection functional

What actually chooses among the seven is not taste but priority order. Give the ten obligations a lexical ordering and maximize co-adaptation subject to everything above it, and each archetype appears as the optimum of a different ordering: put price efficiency first and B follows; learning velocity first and C follows; coherence first and F; neutrality first and G; auditability first and E. The Fabric is what falls out of exactly one commitment: $\{P6, P8\} > \{P3, P10\} > \text{efficiency}$ — sovereignty and honest time are lexical, co-adaptation is then maximized subject to them, and efficiency spends whatever freedom remains. There are $10! = 3,628,800$ strict priority orderings, but they collapse onto this same handful of basins — which is the deep reason “how many other ways?” has the answer it does: **combinatorially many syntheses, and only about seven kinds**. Choosing an architecture is choosing which obligations may never be traded away; everything else in this thesis is consequence.

8.6 Threats to this analysis

The stance enumeration is one designer's partition of a continuous space; a different analyst might split or merge stances and obtain different raw counts. The compatibility rules encode judgments (e.g., that licensing collapses agent markets into crews) that are arguable at the margin. The archetype quotient is robust to these choices in kind but not in number: the honest claim is that the admissible space collapses to *order ten* families, not to exactly ten. None of these threats disturbs the chapter's method — enumerate, prune, quotient, select — which is offered as reusable machinery for design-space argument generally.

9 · Implementation Roadmap and Validation Programme

Five phases, ordered by the governing rule: substrate physics first, one sovereign estate second, federation third, learning-federation fourth, generational maturity last. Every phase carries explicit build steps, a validation burden, and an exit criterion.

9.1 Phase 0 — Substrate and calculus (foundation)

Steps. (1) Implement SB-4, the provenance chain, first — nothing else may come online unwitnessed. (2) Implement SB-2, the λ auction, with the conservative rounding law built into its estimators. (3) Assemble SB-1, the anytime kernel library, rejecting any algorithm that cannot prove interruptibility and monotone answer quality. (4) Implement the spine skeleton SP-1...SP-5 against a stub estate: contracts, gates, refusal with counter-offer, stamps. (5) Define the trace grammar (the six mandatory fields of ES-4) as a wire standard. (6) Stand up SB-3, the twin world, as a replay harness over synthetic workloads. (7) Freeze SB-5, the loom protocol, as the versioned boundary all later work programs against.

Validation burden. Adversarial latency injection cannot make a gate silently late (I2 holds under fault); the auction starves ambiguous bids (safe-side rounding demonstrated); the provenance chain survives third-party audit replay. **Exit criterion.** A stub query descends, is refused honestly at an unsatisfiable contract, and both outcomes appear on the chain.

9.2 Phase 1 — One sovereign estate (single-tenant operation)

Steps. (1) Erect ES-0 boundary and key ceremony; the client holds K_T from the first hour. (2) Load ES-1 vault; implement curation verbs. (3) Build strata S0/S1/S2 and bind them to rungs R0-R2. (4) License the crew charters ES-5; run foragers and curators live, tuners in shadow. (5) Enforce the deposit duty — every search, including refusals, writes a complete trace. (6) Bring ES-3 online with the drift gauge recording and the quarantine cell enforcing between-mission release. (7) Drill ES-6: full secession export to cold storage, verified restorable.

Validation burden. Deposit completeness $\approx 100\%$ under load; quarantine provably interposes on every θ delta (an attempted hot update fails closed); the secede drill completes within its contractual bound and leaves zero residue by independent inspection. **Exit criterion.** Thirty days of single-estate operation with

I1, I2, I4, I6 continuously green on the chain.

9.3 Phase 2 — Federation (coupling without learning)

Steps. (1) Onboard estates two and three under identical constitutions. (2) Open C0 pulse with ε -meters live from the first byte. (3) Stand up MB-5 contribution accounts in shadow. (4) Open C1 void-maps behind incubators; measure forage-efficiency lift attributable to shared negative knowledge. (5) Rehearse MB-6 revocation quarterly: close a live consent, verify drain semantics and meter freeze. (6) Extend R3 campaign contracts to consented cross-tenant recall.

Validation burden. ε accounting exact under audit replay; revocation drains within bound; no cross-estate influence path exists outside the membrane rows of the registry (verified by information-flow test). **Exit criterion.** Demonstrated C1 benefit with bounded, metered leakage — the empirical warrant I3 requires before any wider channel opens.

9.4 Phase 3 — Learning federation (the pipeline, gated)

Steps. (1) Open C2 gradient federation under secure aggregation, incubators mandatory. (2) Bring EV-1 distillery online over consented ledgers; publish regret spectra to tenants. (3) Activate EV-2...EV-4 against the twin world only. (4) Seat the council formally (CH-1...CH-3); run EV-5 on the first surviving proposals. (5) Ship the first micro-generation through EV-6 — a deliberately small change (one API verb, one price revision) to exercise rollout and rollback rails end-to-end.

Validation burden. Every shipped mutation cites traces (I5 audit); the gauntlet demonstrates teeth by refuting a majority of first-round proposals; rollback executes within bound on a staged failure. **Exit criterion.** One full experience→design cycle on the chain, dissent register non-empty (a council that has never dissented has not yet governed).

9.5 Phase 4 — Generational maturity (co-adaptation under damping)

Steps. (1) Move to a regular generation cadence (weeks). (2) Bring EV-7 damper estimation live; publish loop-gain estimates with every release. (3) Open C3 consortium shards for tenants whose contribution accounts support them. (4) Operationalize the fossil record's return rule — a rejected proposal re-enters only against its killing argument. (5) Run annual secede-and-restore drills per tenant; publish results.

Validation burden. Damper gain measurably below one across three consecutive generations; refusal rates at fixed floors non-increasing generation over

generation (the design loop is reducing, not gaming, regret [24]); tail-event retention demonstrated under budget pressure. **Exit criterion.** None — this is the operating state. Maturity (P1) is not a milestone but the condition in which every prior burden stays continuously discharged.

10 · Discussion

Answers to the research questions, the observations that would falsify the design, and the limitations of a notional synthesis.

10.1 Answering the research questions

RQ1 is answered by the ten-obligation decomposition and its traceability map (Chapter 3): every clause of the brief acquires a named discharge site. **RQ2** is answered by the architecture of Chapters 4–7, with ambiguity controlled by three devices: component-before-relationship ordering, the closed-world Interface Registry, and the invariant lattice with named enforcement loci. **RQ3** is answered quantitatively in Chapter 8: 589,824 raw stance-combinations, roughly 1.2×10^5 jointly admissible, approximately ten equivalence families of which seven are load-bearing, and a lexical selection functional under which the Fabric is the optimum. **RQ4** is answered by the phased programme of Chapter 9, whose validation burdens are pre-registered claims, and by the falsifiers below.

10.2 Falsifiability

Four falsifiers, stated against the thesis’s own claims. **First**, if metered coupling never demonstrates benefit — if C1/C2 lift over a C0-only baseline is statistically indistinguishable from zero in twin-world replay — then the membrane is apparatus without payoff and the fabric degenerates, honorably, into the Archipelago. **Second**, if the resonance damper must hold loop gain far below one to keep the federation stable, then generational co-adaptation is theater: the system is safe only because it barely co-adapts, and archetype E was the honest choice. **Third**, if refusal rates at fixed floors rise generation over generation, the design loop is optimizing something other than the regret it cites — Goodhart at the forge [24, 32] — and I5’s citation discipline has failed its purpose. **Fourth**, if a secede drill cannot complete within its contractual bound, sovereignty is decorative and the architecture’s central boast is false. Each falsifier is measurable inside the Phase 2–4 validation programme; none requires waiting for a disaster.

10.3 Limitations and threats to validity

Notional status. No implementation exists; every quantitative claim is arithmetic or pre-registration, not measurement. Design-science evaluation at this stage is analytical by necessity [15], and the thesis has tried to convert that weakness into discipline by pre-committing its validation burdens. **Single-designer bias.** The stance enumeration, compatibility rules, and equivalence quotient of Chapter 8 encode one designer’s judgment; Section 8.6 bounds the damage but cannot remove it. **Apparatus weight.** The Fabric buys its forbidden corner with governance machinery — councils, meters, incubators, twins, gauntlets — whose operating cost is unmodelled; a rival could concede the corner and win on lightness. **Mechanism maturity mismatch.** Adopted components differ in readiness (secure aggregation is production-grade; online learning-coefficient estimation is research-grade [19]), so the roadmap’s Phase 3-4 carry the largest scientific risk. **Socio-technical scope.** The charter plane assumes clients willing and competent to govern; the design does not model council capture, collusion between tenants, or regulatory conflict beyond citing the applicable data-protection regime [10].

10.4 Kinship

The fabric deliberately imports a metabolic doctrine developed in a companion internal specification for bounded autonomous systems — residence-time stratification, shadow-priced recall, drift quarantine, honest refusal — and extends it from a single system’s memory to a federation’s shared habitat, where verification burden grows with a change’s blast radius. The extension is not cosmetic: what changes at federation scale is that the entity bearing a corrupted datum’s blast radius may not be the entity that admitted it, which is why the membrane’s consent-and-meter machinery, absent in the single-system doctrine, carries so much of this design’s weight.

11 · Conclusion and Future Work

What was built, what it claims, and what should be done next.

11.1 Conclusion

This thesis set out to answer one sentence with one architecture. The sentence demanded ten things at once; the architecture delivers them as four planes, two transverse structures, twenty-nine interfaces, and seven invariants: search operated entirely by agent crews inside estates their clients own down to keys and exit; every query admitted under an explicit deadline contract and refused aloud, with a counter-offer, when the contract cannot be honored; tenants coupled through a graded, consented, metered, unilaterally revocable membrane and never merged; an operator reduced to protocol — the loom, never the thread; and a design that is itself a signed, versioned artifact which only recorded experience, distilled and tried and refuted and countersigned, may rewrite, under a damper that keeps co-adaptation below resonance. The design-space analysis shows this is one of about seven coherent ways to keep the brief's promises, and the selection functional makes the choice reproducible: hold sovereignty and honest time lexical, maximize co-adaptation subject to them, and the Fabric is what remains.

11.2 Future work

Four lines follow directly. **Build Phase 0.** The substrate-and-calculus phase is small enough for a single engineering effort and would convert the spine's refusal semantics and the auction's safe-side rounding from design into measurement. **Formalize the damper.** The loop-gain criterion is stated in control-theoretic vocabulary [1] but needs an estimator with known bias under the fabric's sampling regime — the clearest open theory problem in the design. **Calibrate the design-space method.** Chapter 8's enumerate-prune-quotient machinery should be exercised by independent analysts on the same brief to measure the stability of the archetype quotient. **Standardize the trace grammar.** The six-field trace and the stamped-answer format are the fabric's natural first standards; wire-level specifications would let heterogeneous crews and estates interoperate before any full implementation exists.

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Appendix A — Glossary

Table A.1: Terms of art used throughout the thesis.

Term	Definition
anytime kernel	an algorithm that can be interrupted at any moment and return its best answer so far, with quality monotone in time spent (SB-1)
archetype family	an equivalence class of stance-vectors that survive compatibility pruning and differ only in inessential structure (Chapter 8)
charter instrument	the signed tenant document fixing budgets B , floors $q_{\min}$, and consents κ (CH-2)
consent key κ	the live, charter-issued credential that opens a membrane valve between two tenants (MB-2)
contribution account	Shapley-metered give/take balance per tenant pair and channel (MB-5)
counter-offer (d', q')	the nearest honorable contract returned with every refusal (SP-4)
deposit duty	the unconditional obligation that every search, including refusals, writes a complete trace (ES-4, I4)
distillate	a decision-grade compression of ledger traces: regret spectra, friction maps, refusal statistics (EV-1)
drift gauge	the online estimator $(\lambda\text{-hat})$ of model learning-capacity whose spikes arm quarantine (ES-3a)
estate	one tenant's sovereign territory: vault, strata, organ, ledger, crew, ports (Plane 1)
fast loop	seconds-to-hours strategy adaptation by the crew; never touches design (ES-7)
fossil record	the legible archive of refuted proposals and retired organs (EV-8)
generation frame $G(n)$	the versioned, signed design artifact in force; the object the pipeline rewrites (EV-9)
honest refusal	declining a contract aloud, with counter-offer, rather than being silently late or shallow (SP-4, I2)
incubator	the receiving-side quarantine chamber on every membrane channel (MB-4)
loom protocol	the operator's complete external surface: formats, grammars, proof obligations (SB-5)
loop gain	the product of design-to-experience and experience-to-design sensitivities; released designs require it below one (EV-7, I7)
quarantine cell	the holding stage through which every model delta passes between missions (ES-3b, I6)
residence time τ	how long an artifact remains action-relevant; the stratification variable of ES-2
secede port	the verified unilateral exit: full signed export, revoked couplings, zero residue (ES-6, I1)
shadow price $\lambda(t)$	the single posted price rationing all claims on shared capacity (SB-2)
stance	one defensible architectural position on one of the ten obligations (Section 8.2)
trace	the six-field record a search deposits: contract, path, cost, outcome, regret, dead-ends/reformulations (ES-4)
twin world	the shadow replica in which candidate designs replay recorded workloads (SB-3)
void-map	shared negative knowledge: where searching has been shown futile (MB-1b)